When Time Runs Out, Scientists Test a Nuclear Plan to Stop an Asteroid

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Imagine a space rock so big that no gentle nudge could move it in time. That is the puzzle scientists are now testing with computer **simulations**, not real explosions. About 66 million years ago, an asteroid about 6 to 9 miles across, called Chicxulub, struck Earth and wiped out about 75 percent of life, including all **non-avian** dinosaurs. Nothing that large has hit since, but researchers say another big impact is not impossible. In 2022, NASA's DART mission showed one safer idea can work. It struck Dimorphos, the small moon of the asteroid Didymos, and changed its orbit. Didymos was small, however, and was never a threat. For a much larger rock with very little warning time, launching many **impactors** would be impractical. So a team at Lawrence Livermore National Laboratory, led by physicist Isaiah Santisevan, simulated a nuclear explosive device instead. In the most common model, called a **standoff burst**, the device explodes above the asteroid. Most of its energy arrives as X-rays, which heat a thin surface layer until it vaporizes. That material rushes outward, pushing the asteroid the other way like a rocket push. The blast can also send a shock wave through the rock and break it into smaller pieces. The team ran 1,500 one-dimensional **simulations**, then moved to three-dimensional models of an asteroid about 525 feet across, changing the detonation distance and the rock's material. In the longest simulation, the asteroid broke apart, and most of its material moved in the opposite direction, faster than the asteroid's **escape velocity**. Future work will track where those fragments would go. The researchers describe the results as a step toward solving an immensely difficult problem in planetary defense.

Word Treasure

simulations -- Computer models that act out what might happen in real life.

non-avian -- Dinosaurs that are not birds.

impactors -- Spacecraft sent to crash into an asteroid to push it.

standoff burst -- An explosion set off above an asteroid, not on its surface.

escape velocity -- The speed something needs to escape an asteroid's gravity and never fall back.

Background you need

An asteroid is a rocky body that orbits the Sun, and a very large one hitting Earth could cause damage over a huge area.

Planetary defense means finding ways to change such a rock's path or break it up before it can hit Earth.

One tested idea is to crash a spacecraft into the rock, which can nudge its orbit, but that takes enough warning time and a rock that is not too large.

A nuclear device set off above the rock is another idea being studied in computer models, where heat from the blast pushes on the surface and moves the rock.

Break it down

LEAD: Scientists test a nuclear plan in computer simulations for a huge asteroid with little warning time.

BACKGROUND: About 66 million years ago, Chicxulub wiped out about 75 percent of life, and in 2022 DART changed a small asteroid moon's orbit.

KEY EVENT: A Lawrence Livermore team simulates a nuclear device exploding above an asteroid.

+- HOW: X-rays heat a thin surface layer until it vaporizes, pushing the asteroid the other way.

+- EVIDENCE: 1,500 one-dimensional simulations, then three-dimensional models of a 525-foot asteroid.

OPEN QUESTION: Where would the fragments go?

Why it matters

A giant asteroid strike is rare but could harm the whole planet, so finding a workable last-resort defense matters for everyone's future safety.

Test what you remember

1. **About how wide was the asteroid Chicxulub?**
   - (A) About 6 to 9 miles
   - (B) About 1,500 miles
   - (C) About 525 feet
   - (D) About 75 miles
2. **What did NASA's DART mission do in 2022?**
   - (A) It struck Dimorphos and changed its orbit
   - (B) It exploded above Dimorphos
   - (C) It moved Dimorphos to a new star
   - (D) It broke Dimorphos into pieces
3. **Why would launching many impactors be impractical for a very large asteroid with little warning time?**
   - (A) Impactors only work on small moons
   - (B) The impactors would melt in space
   - (C) The rock would be too far away to see
   - (D) There would not be enough time to launch enough of them
4. **In a standoff burst, where does the nuclear device explode?**
   - (A) Inside the asteroid's core
   - (B) On the far side of the Moon
   - (C) Above the asteroid
   - (D) Under the asteroid's surface
5. **What do the X-rays do in the standoff burst model?**
   - (A) They light up the sky for telescopes
   - (B) They freeze the asteroid's surface
   - (C) They heat a thin surface layer until it vaporizes
   - (D) They send a radio signal to Earth
6. **What happened in the team's longest simulation?**
   - (A) The asteroid stayed whole and sped up
   - (B) The asteroid broke apart and most of its material moved faster than its escape velocity
   - (C) The asteroid turned into a comet
   - (D) The asteroid fell back to Earth

Answer key (try the quiz first!): 1: A 2: A 3: D 4: C 5: C 6: B

Questions to think about

1. Researchers: Say another big impact is not impossible and describe the results as a step toward solving an immensely difficult problem in planetary defense.

MY THOUGHTS (at least 20 words):